

Bi-Clustering Methods

- ❑ Real-world data is noisy: Try to find approximate bi-clusters
- ❑ Methods: Optimization-based methods vs. enumeration methods
- ❑ Optimization-based methods
 - ❑ Try to find a submatrix at a time that achieves the best significance as a bi-cluster
 - ❑ Due to the cost in computation, greedy search is employed to find local optimal bi-clusters
 - ❑ Ex. δ -Cluster Algorithm (Cheng and Church, ISMB'2000)
- ❑ Enumeration methods
 - ❑ Use a tolerance threshold to specify the degree of noise allowed in the bi-clusters to be mined
 - ❑ Then try to enumerate all submatrices as bi-clusters that satisfy the requirements
 - ❑ Ex. δ -pCluster Algorithm (H. Wang et al.' SIGMOD'2002, MaPle: Pei et al., ICDM'2003)

Bi-Clustering for Micro-Array Data Analysis

- ❑ Left figure: Micro-array “raw” data shows 3 genes and their values in a multi-D space: Difficult to find their patterns
- ❑ Right two: Some subsets of dimensions form nice shift and scaling patterns
- ❑ No globally defined similarity/distance measure
- ❑ Clusters may not be exclusive
 - ❑ An object can appear in multiple clusters

